# PROJECT ID:- 2026-MPJ-CSDS-4
# "SMART MOVE"

## Department of Computer Science & Engineering (Data Science)


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# ***Objective of Project***



- To develop a low cost and user-friendly real time public transport tracking system for small cities.
- To provide passengers with live transport locations, routes,stops, and estimated arrival times.
- To enable drivers to manage trips and admins to monitor bus status.
- To improve public transport convenience through accurate and updated bus information.

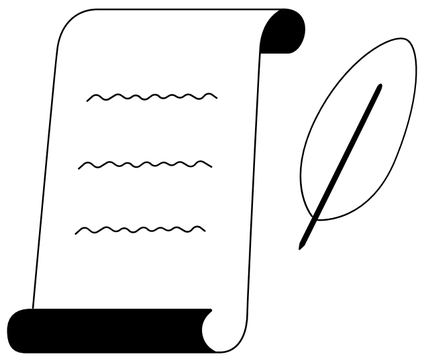
# *Problem Statement*

***"REAL-TIME PUBLIC TRANSPORT TRACKING FOR SMALL CITIES"***

- No real-time transport tracking in many small cities.
- Passengers face long waiting times.
- No accurate arrival/ETA information.
- Difficulty in finding suitable transport.
- Lack of efficient transport status & trip management.
- Limited access to reliable & updated transport information.



# ***Introduction***



- Introduce the need for smart public transport.
- GPS-based system tracks transport in real time.
- Passengers can find transport using source and destination
- Live transport location and ETA improve travel planning
- Drivers & Admins can manage trips and transport status
- Affordable & Scalable solution for small cities & towns.

# *Literature Survey*

| AUTHOR & YEAR | TECHNOLOGY/METHOD | MAJOR CONTRIBUTION |
|---|---|---|
| Ganesh et al. (2012) | GPS, GPRS, ETA, Algorithm, Web Application | Developed a real-time passenger information system providing bus locations and estimated arrival times. |
| Bojan et al. (2014) | IoT, GPS , Sensors | Proposed an IoT based system for monitoring vehicle location, passengers and transport conditions. |
| Varma et al. (2024) | GPS, Machine learning, Kalman Filter, SVR, KNN | Developed a real-time bus tracking system with machine learning based location and arrival-time prediction. |
| Anuprathibha et al. (2025) | Sensor Technology, Real-Time Tracking | Proposed real-time public transport tracking to overcome limitatioons of fixed schedules and manual reporting. |

# *Proposed Methodology*

- GPS-based location tracking
- Route & stop mapping
- Real-time data processing
- ETA Estimation
- DrIver authentication & trip control
- Passenger bus search
- Bus status management
- Location permission & GPS checks

# Proposed Implementation

**Frontend:** React/HTML-CSS-javascript-node.js

**Backend:** python/Node.js

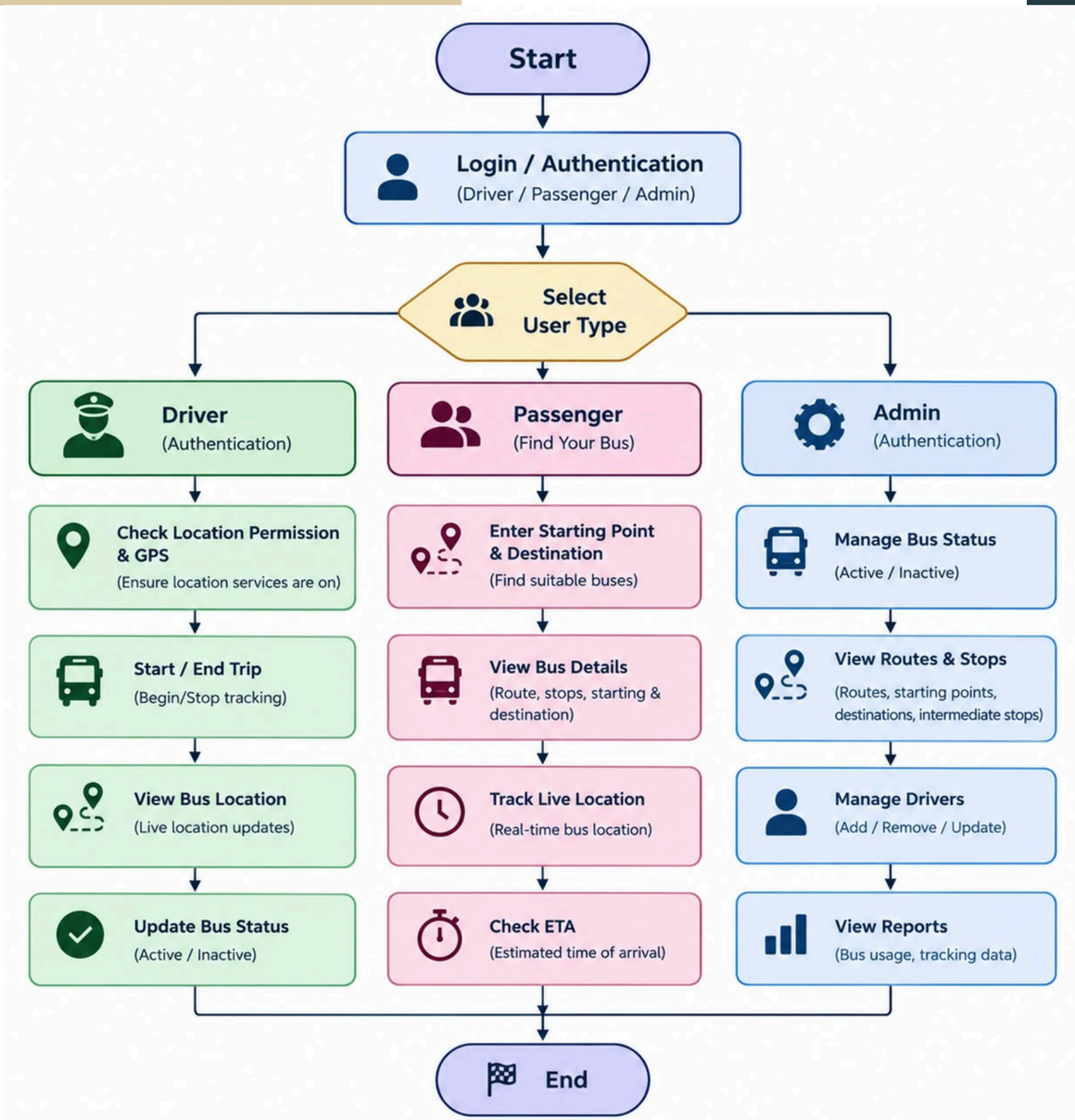
**Database:** Firebase/SQL

**Maps:** Open street Map/ google Maps

**Location:** smartphone GPS

**Modules:** Passenger + Driver + Admin dashboard

## WORKING:



# *Reference*


- **Ganesh et al.(2012)-**implementation of real time passenger information system.
- **Bojan et al.(2014)-**An internet of things based intelligent transportation system.
- **Varma et al(2024)-**Machine learning enabled smart transit.
- **Anuprathibha et al.(2025)-**Real time tracking of public transport using sensor Technology.
- **Google maps/openstreetmap documentation-** Mapping and Location services
- **Firebase Documentation-** Real-time database and authentication .
- **GTFS Realtime Documentation-** Public transport real-time data standards

***Thank you!***